Eli Coe Naig

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Education

Tulane University, School of Science and Engineering — New Orleans, LA
B.S. in Mathematics, GPA: 3.97 / 4.00 — August 2024 – May 2027, Expected

Research Interests

Post-quantum and code-based cryptography for resource-constrained devices; algebraic coding theory; hardware-assisted key establishment; quantum information theory.

Publications, Preprints & Manuscripts

[1] M. B. Can and E. C. Naig, "A Theory of Leakage in Secret Sharing and Evaluation Codes", Submitted, under review, 2026.

[2] M. B. Can and E. C. Naig, *Applications of Bruhat-Chevalley-Renner Decomposition to Metric-Aware Code-Based Cryptography*, Cryptology ePrint Archive, Paper 2026/665, Preprint; submitted to the Journal of Cryptology, 2026. [Online]. Available: https://eprint.iacr.org/2026/665

[3] M. B. Can and E. C. Naig, "Module-HQC", Submitted to IEEE Transactions on Information Theory, 2026.

[4] E. C. Naig, M. B. Can, and S. Chakrabartty, "On Timekeeping-Based Key Establishment Protocols", Submitted to IEEE Transactions on Information Forensics and Security, 2026.

[5] E. C. Naig and S. A. Fenner, *Parallel Repetition in the Two-Player Quantum Cloning Game*, arXiv:2608.00930 [quant-ph], Preprint, 2026. [Online]. Available: https://arxiv.org/abs/2608.00930

Talks & Presentations

- *Parallel Repetition in the Two-Player Quantum Cloning Game.* Talk presented by co-author S. A. Fenner, Ulm University, Ulm, Germany (event page). September 2026
- *Quantum Position Verification and the Quantum Cloning Game.* Poster, USC Summer Research Symposium, University of South Carolina. July 2026
- *Quantum Position Verification: Cloning Games and an Open Repetition Problem.* Talk to faculty and students of the NSF REU in Quantum Information Science, University of South Carolina. July 2026

Patents

- **Tunable Module Hamming Quasi-Cyclic Code-Based Cryptography**. U.S. provisional patent application, with M. B. Can. Filed August 2026
- **Systems and Methods for Secure and Reliable Data Storage and Transmission**. PCT patent application, with M. B. Can and D. Montero. Filed April 2026
- **Metric-Aware Code-Based Cryptography**. U.S. provisional patent application, with M. B. Can. Filed March 2026

Research Experience

Undergraduate Researcher — New Orleans, LA
Advisor: Mahir Bilen Can, Tulane University — August 2025 – Present

NSF REU in Quantum Information Science — Columbia, SC
Advisor: Stephen Fenner, University of South Carolina — May 2026 – August 2026

RTG Summer Math Program — Baton Rouge, LA
Louisiana State University — May 2025 – June 2025

Technical & Entrepreneurial Experience

QSPARC Labs — New Orleans, LA
Co-Founder, CEO, and Lead Developer — August 2025 – Present

- **Open-Source Software**: Contributor to RustCrypto and SageMath; member of PassageMath.

Teaching & Mentoring

- Volunteer Tutor, Excite All Stars — Fall 2026 – Present
- Volunteer College Application Coach, College Beyond — Fall 2026 – Present
- Volunteer Tutor, Tulane University — Fall 2025 – Present

Awards & Grants

- Tulane Honors Scholar — 2024–Present
- Dean's List, Tulane University — Fall 2024; Spring 2025; Fall 2025; Spring 2026
- CELT Faculty-Mentored Undergraduate Research Grant, Tulane University; mentor: Mahir Bilen Can. — 2026
- Chosen for a Tulane Innovation Institute program on research translation and proposal development. — 2025
- Ellen Applebaum Temkin Scholarship, Mathematics Department, West High School — 2024

Leadership & Service

- President, Tulane Math Club — Spring 2026–Present
- Co-Founder and Vice President, Tulane Science Olympiad — 2024–Present
- Trip Leader, Tulane Outdoor Adventures — 2024–Present